

What Comes from the Agent, and What Comes from the Rules?

A Study of Two LLM Protocols

Abstract

The pair consists of a study of language-model traders in a continuous double auction and a study of adversarial review by language-model coding agents. We looked for a common way to separate the value of agent information from the effect of the rules under which agents interact. For the market paper, two independent simulations found that constrained random traders reach about 0.96 allocative efficiency, above every reported language-model result. For the review paper, the available results do not separate the critic’s information from repeated review and the rule for stopping. The thread therefore paused because it found a sound result for the market paper but only an untested connexion between the papers.

1 Introduction

Struski and colleagues study language models as traders in a continuous double auction [5]. Buyers and sellers repeatedly post prices, and a trade occurs when a bid meets an ask. The paper asks whether a market mechanism that works well with people also produces efficient allocations when the traders are language models. Its reported language-model markets converge more slowly than the human comparison, or do not converge, and their allocative efficiency ranges from 0.36 to 0.91.

Qiu and Gill study Adversarial Review, or AR, for coding tasks [4]. A main agent writes code, a reviewer assesses it, and a critic checks the review. The reviewer and critic exchange structured judgements before the main agent edits the code. The paper reports improvements on several benchmarks, but also describes false consensus: the reviewer and critic can agree without enough evidence.

The connexion pursued was institutional. In both settings, an observed outcome can come from the agents’ information, or from the rules governing allowed messages, repeated attempts, and stopping. A useful control should hold the institution fixed while removing informed behaviour. The market setting admits constrained zero-intelligence traders, known as ZI-C traders [3]. An

analogous review control would replace the critic’s verdict with a random verdict while preserving the rest of the review loop.

2 What was done

The peers first rebuilt the market described by Struski and colleagues. Each round has 11 buyers and 11 sellers. An agent is selected at random to act, the order book persists, and a round permits at most 300 selections. A ZI-C buyer quotes a random price no higher than its private value. A ZI-C seller quotes a random price no lower than its private cost. Thus the trader has no strategy, but it does not knowingly accept a loss. The two implementations and their outputs are in `ada/cda_sim.py`, `ada/cda_sim_out.txt`, `emmy/sim_cda.py`, and `emmy/sim_results.json`.

The peers then tested whether the comparison depended on implementation details. They varied whether standing quotes survived a trade and varied the range from which random prices were drawn. They also ran the random market for shorter horizons. This gave a “ZI-equivalent horizon”: the number of random selections needed to match each language model’s mean efficiency.

Next, the peers split efficiency loss into two kinds. Let E be allocative efficiency, meaning realised surplus divided by the maximum feasible surplus. Let O be the share of maximum surplus lost because profitable traders were left out, and let C be the share lost because unprofitable, extramarginal traders took part. In this market the accounting identity is

$$1 - E = O + C. \tag{1}$$

Here “omission” means O and “commission” means C . They also tested simple rule-based traders that moved prices by one cent, and traders that accepted any individually profitable standing quote. These were illustrative models, not models of the language-model traders.

The work then turned to the review paper. The peers compared its methods by what happens after an edit. Single-reviewer and Two-reviewers review once and then edit once. MARS and AR review a revised artefact again. These latter methods are also the methods that rise above the 75 to 77 per cent cluster in the paper’s LiveCodeBench results. The peers therefore built a small probability model to check whether iteration alone could matter. The code and output are in `ada/review_loop_toy.py` and `ada/review_loop_toy_out.txt`. This model was a logical check only; it did not use the review paper’s task data.

Finally, the peers inspected the market paper’s released repository [2] and checked both papers’ stated procedures. They narrowed claims whenever the code or experimental design did not support the first interpretation. A search also found BAZAAR [1], but it studies a sealed-bid double auction and reports individual profit rather than market efficiency. It therefore did not supply a close comparison or a novelty claim.

3 Results

The firm result concerns the market paper. The two independent ZI-C implementations produced efficiency between 0.956 and 0.961, about 5.8 to 5.9 trades per round, and a mean transaction price of 2.00. The second implementation used 2,000 rounds per configuration. Varying the random quote range left efficiency between 0.943 and 0.957. These checks support a ZI-C benchmark near 0.96 for the stated mechanism.

Every reported language-model cell in the market paper lies below this benchmark, at 0.36 to 0.91. The difference is not a claim that random traders are generally better traders. It says that, in this mechanism and under the constraint just defined, random quotes extract more of the available surplus than the reported language-model runs.

This horizon measure makes the gap easier to read. Although each language-model round allows 300 selections, ZI-C reaches the five-round mean efficiency of GPT Large after about 53 selections, Gemini Large after about 80, and GPT Small after about 131. The supporting calculation and output are in `ada/cda_tick.py` and `ada/cda_tick_out.txt`.

The surplus split also holds exactly for the simulated market. Maximum feasible surplus is 7.50. The five largest efficient buyer-seller pair surpluses are 2.50, 2.00, 1.50, 1.00, and 0.50. ZI-C produced $O = 0.009$ and $C = 0.032$. A one-cent rule-based trader mainly omitted profitable trades, while an accept-by-default trader raised volume but produced C between 0.10 and 0.17 and efficiency between 0.71 and 0.79. The independent acceptor check gave similar values. Making trade easier therefore changed the kind of error without recovering ZI-C efficiency.

There is no corresponding firm result for the connexion to AR. The toy review model was calibrated so that one review-and-edit cycle had no net benefit. Under its chosen parameter sets, review until approval added 0.3 to 9.1 percentage points, with a median of 2.6 points. This shows that repeated review can matter in principle. It does not show that repeated review caused the gains in Qiu and Gill’s experiments.

4 What did not hold, and the objections

An early claim treated one cent as an enforced tick size in the market. Inspection of the released code showed otherwise. Language-model orders need only improve a standing quote by a strict floating-point amount. The one-cent increment is used in a deadlock check for zero-intelligence agents, not as a general constraint. The peers therefore retained the tick experiment only as a prediction for a new mechanism with an enforced and stated tick, not as a result about the paper’s actual language-model runs.

The comparison also needs a qualification about private values. The exchange does not enforce buyers’ values or sellers’ costs, although the prompt tells language-model agents never to cross those limits. The code logs viola-

tions, but the paper does not report their rate. The 0.96 figure is therefore a constrained-strategy benchmark conditional on language-model compliance with the prompt, not an unconditional baseline for everything the exchange permits.

The direction of market failure was also narrower than first stated. GPT Large and Gemini Large mainly omit trades. GPT Small in its fifth round, and unreported Gemini runs found in the authors’ notebook, also show commission. The direction depends on the model. In addition, two ZI-C batches in the authors’ notebook have different price-dispersion values, and the available material does not explain the difference.

The main objection concerned the cross-paper claim. AR adds a critic, but it also re-reviews revised code and changes when the process stops. The paper does not include a reviewer-only iterate-until-clean arm or a random-critic arm. It therefore cannot identify the value of critic information separately from re-review, stopping, and adaptive compute. The paper also reports MARS as 82 per cent in its main table but describes it as 85 per cent in prose. That inconsistency remains unresolved.

5 Why the thread stopped

Across all three rounds, the verifier accepted the ZI-C result but asked for a matched AR ablation. The required comparison is reviewer-only iteration until clean, ordinary AR, and AR with a random critic whose disagreement rate matches the observed critic. The same tasks, outer limits, and compute budget must be used, with paired outcomes reported for each task.

That experiment could not be run from the supplied material. The review paper gives aggregate results and prompts, but the workspace contains no task instances, per-task outcomes, review traces, empirical disagreement rates, runnable orchestration, or model credentials. Using the toy model in their place would invent evidence. The smallest step that would restart the thread is therefore access to the original tasks and traces, including per-round critic disagreements, plus an execution environment in which the three matched arms can be run. If ordinary AR beats both controls, the critic’s information has an effect beyond re-review and stopping; until then, the proposed connexion remains an untested causal question.

References

- [1] BAZAAR.
- [2] Competitive market simulation, 2026.
- [3] Gode and Sunder. Zero-intelligence constrained traders, 1993.
- [4] Eric S. Qiu and Joyce Gill. Adversarial review: Structured disagreement for grounded agentic code review. arXiv:2608.18167, 2026.
- [5] Pawel Struski, Jakub Swistak, Inez Okulska, and Przemyslaw Biecek. Competitive market behavior of LLMs. arXiv:2609.02580, 2026.